
Bachelor of Business Administration(BBA)

Course Title: IT 233

Business Information Systems

Fifth Semester

Compiled by SKM

Course Contents

Our Recommended Text books:

1. Management Information Systems: Managing the Digital Firm, Kenneth C. Laudon and Jane P. Laudon, 17th Edition, Pearson, 2022.
2. Management Information Systems, James O'Brien and George Marakas, 10th Edition, McGraw Hill.

Course Contents

Unit 1: Introduction to Information Systems(3 Hrs.)

- Introduction
- Why should we study information systems?
- Computer-Based Information Systems and Types
- Impact of IT on Organizations
- Importance of Information Systems to Society

Introduction to Information Systems

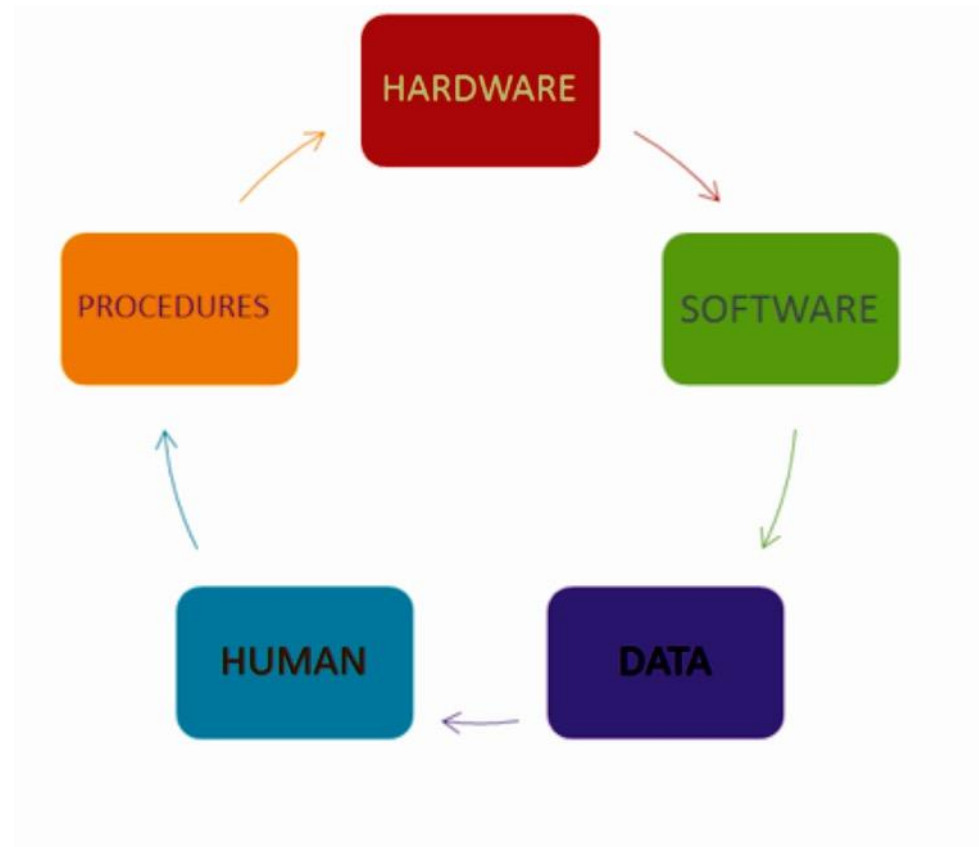
An Information system that is the sub system of the business. Especially, it is an arrangement of interdependent of human and machine components that interact to support the operation when organization needed decision making. In other words, An information system (IS) is an interconnected set of components used to collect, store, process and transmit data and digital information.

Introduction to Information Systems

The components or resources of Information systems are

- People or human
- Hardware
- Software
- Data
- People or human &
- Procedures

Resources of Information System



Why should we study Information systems?

- Multiple career paths
- Flexibility
- Career advancement
- High demand for skills
- Continue learning
- High earning potential
- Transferable skills
- Creative expression

Computer-Based Information System

A computer-based information system refers to a collection of various systems such as office automation systems, transaction processing systems, management information systems and management support systems that utilize computers to process and manipulate data in order to achieve specific objectives.

Types of Information System

1. Strategic Information System
2. Operational Information System
3. Tactical or MIS

Strategic Information System

This information is required by top most management for long range planning policies for next few years. As for example, financial investment, population growth and human resources etc. A strategic Information system(SIS) is a system that helps companies change their business, strategy and structure.

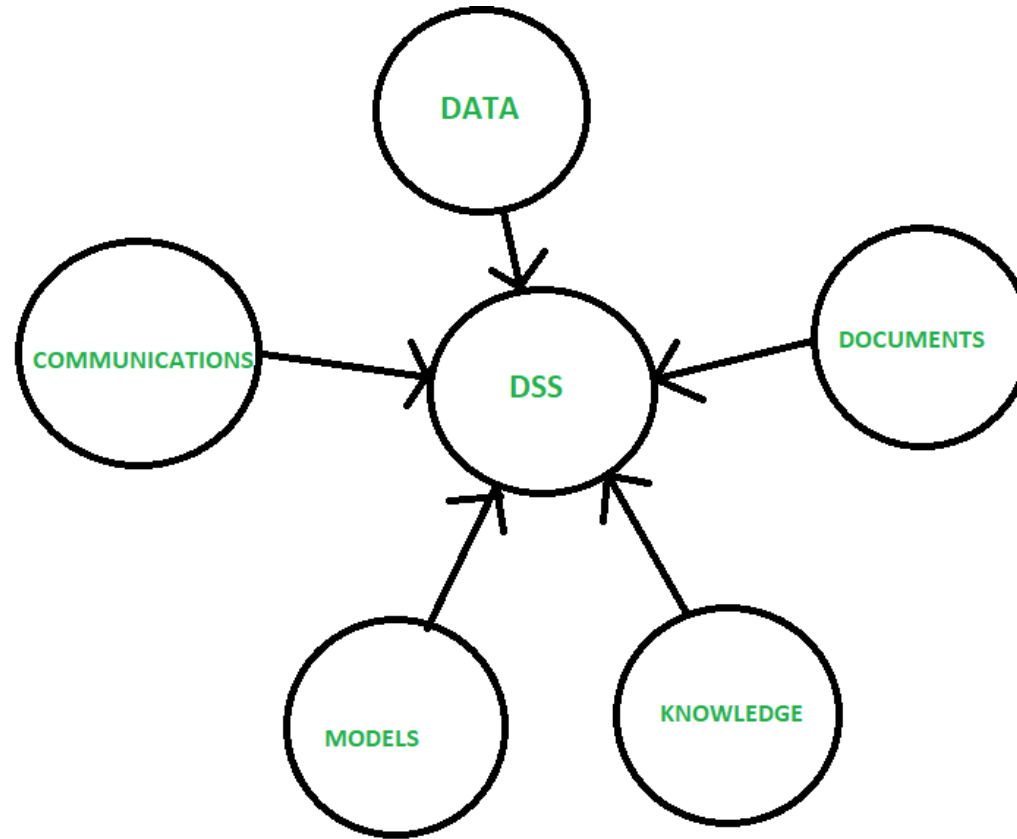
- Decision support system(DSS)
- Artificial Intelligence(AI)
- Expert System(ES)
- Executive Information System(EIS)

Strategic Information System

- Decision support system(DSS)

A DSS is an information system that support business or organizational decision-making activities. A decision support system(DSS) is a computer based application that collects, organizes and analyzes business data to facilitate quality business decision-making for management, operations and planning.

Strategic Information System

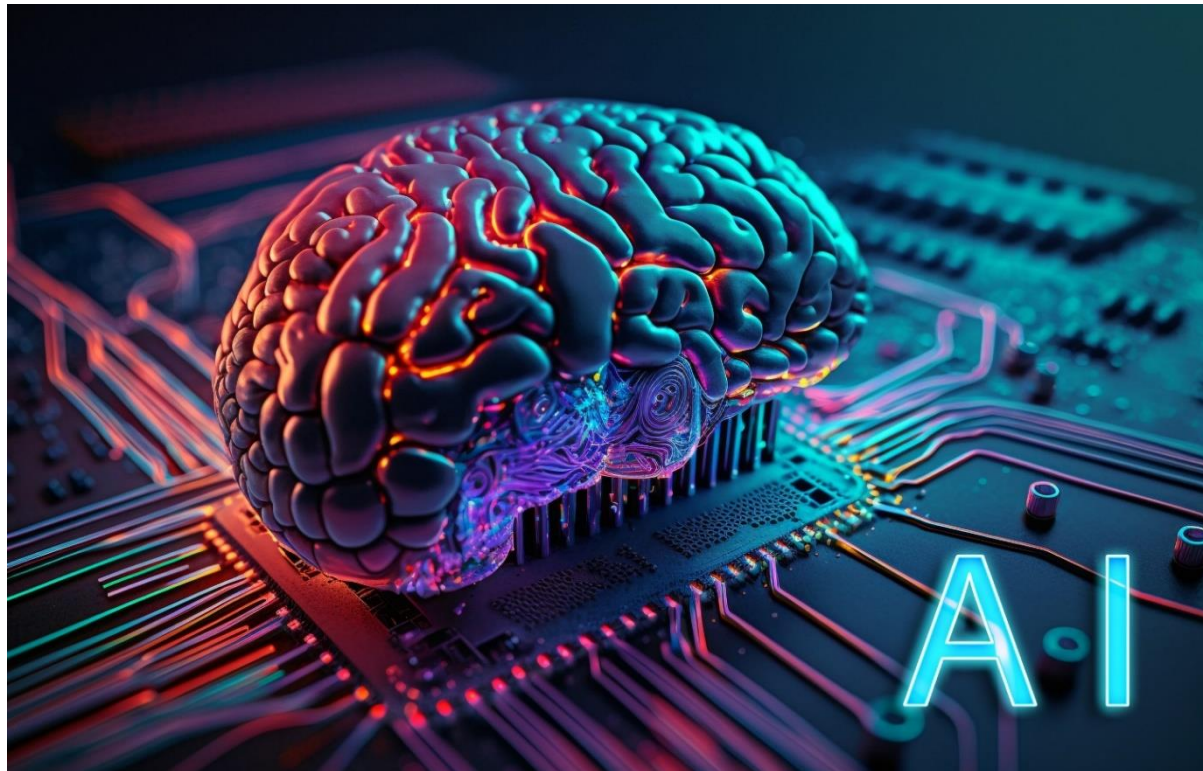


Strategic Information System

- Artificial Intelligence(AI)

AI is the branch of computer science whose goal is to develop a machine with the human intelligence.

Strategic Information System



Features of AI

- Deep learning
- Natural language processing
- Automating repetitive tasks
- Decision-making
- Automation
- Efficiency
- Cloud Computing

Strategic Information System

Four goals of AI

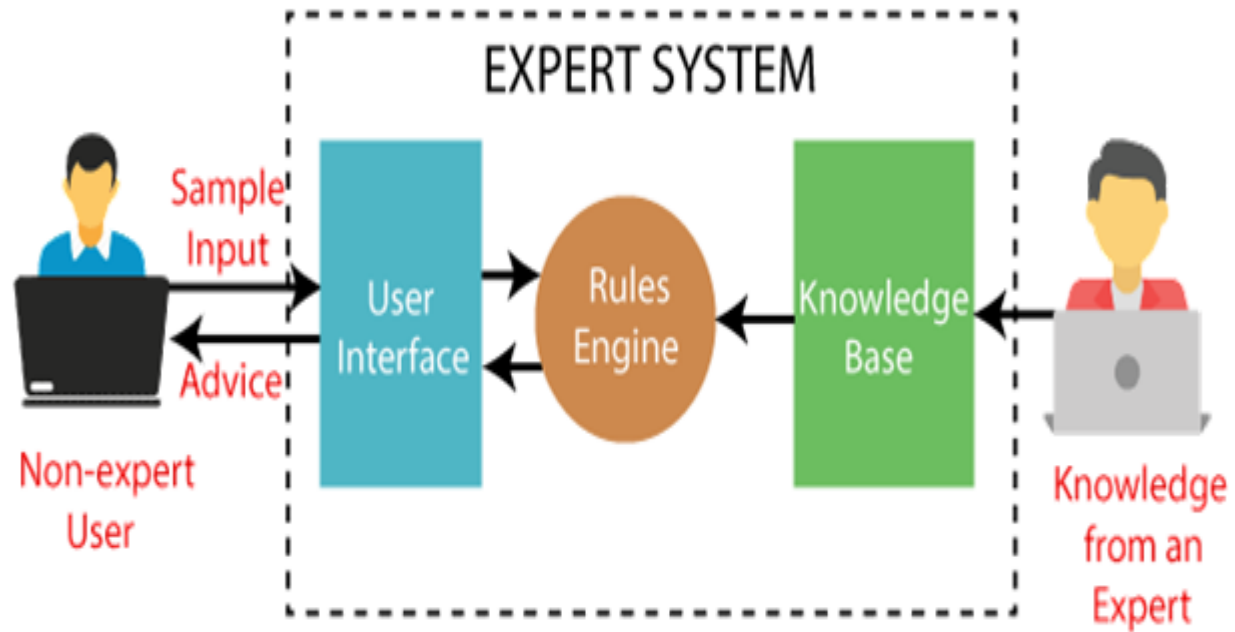
- System think like human.
- System act like human.
- System think rationally.
- System act rationally.

Strategic Information System

- Expert System(ES)

An expert system is a computer based system that uses AI technologies to simulate the judgment and behavior of a human or an organization that has expert knowledge and experience in a particular field.

Strategic Information System

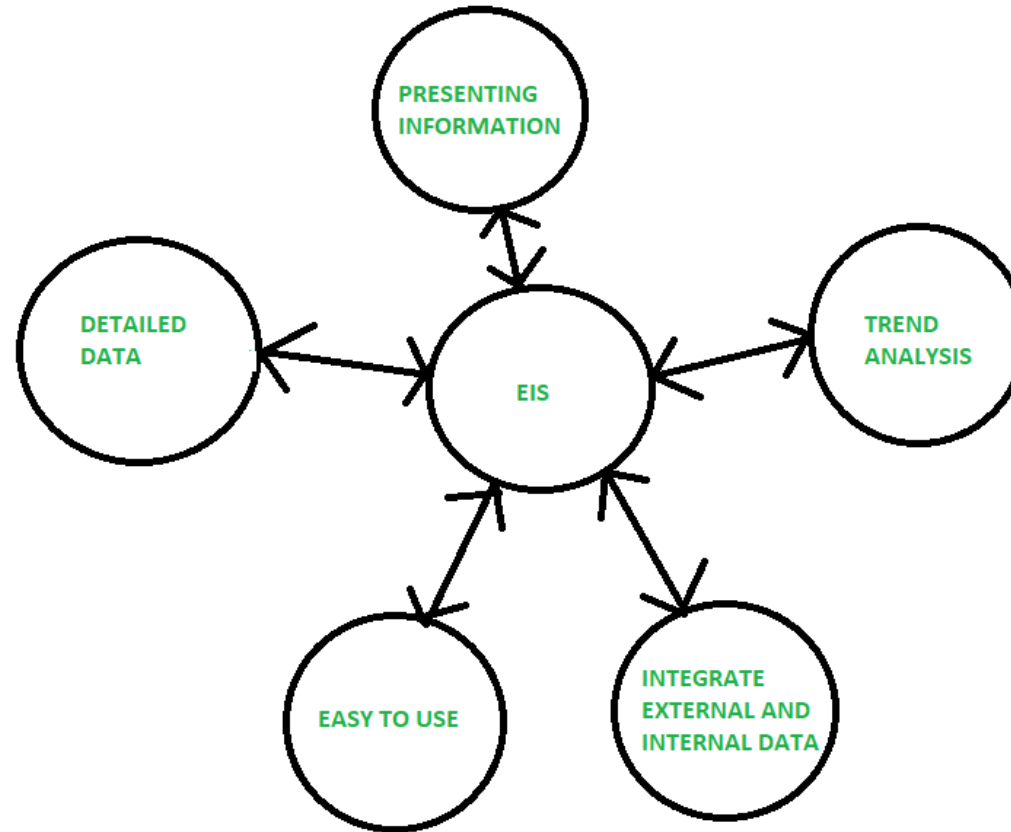


Strategic Information System

- Executive Information System(EIS)

An **executive information system (EIS)** also known as an **executive support system (ESS)** is a type of management support system that facilitates and supports senior executive information and decision-making needs. It provides easy access to internal and external information relevant to organizational goals. It is commonly considered a specialized form of decision support system(DSS).

Strategic Information System



Operational Information System

This type of information is required by low management for daily and short term planning. This system can be used day by day activities of an organization. As for example, keeping employee attendance activities, overdue purchase and current stocks available etc.

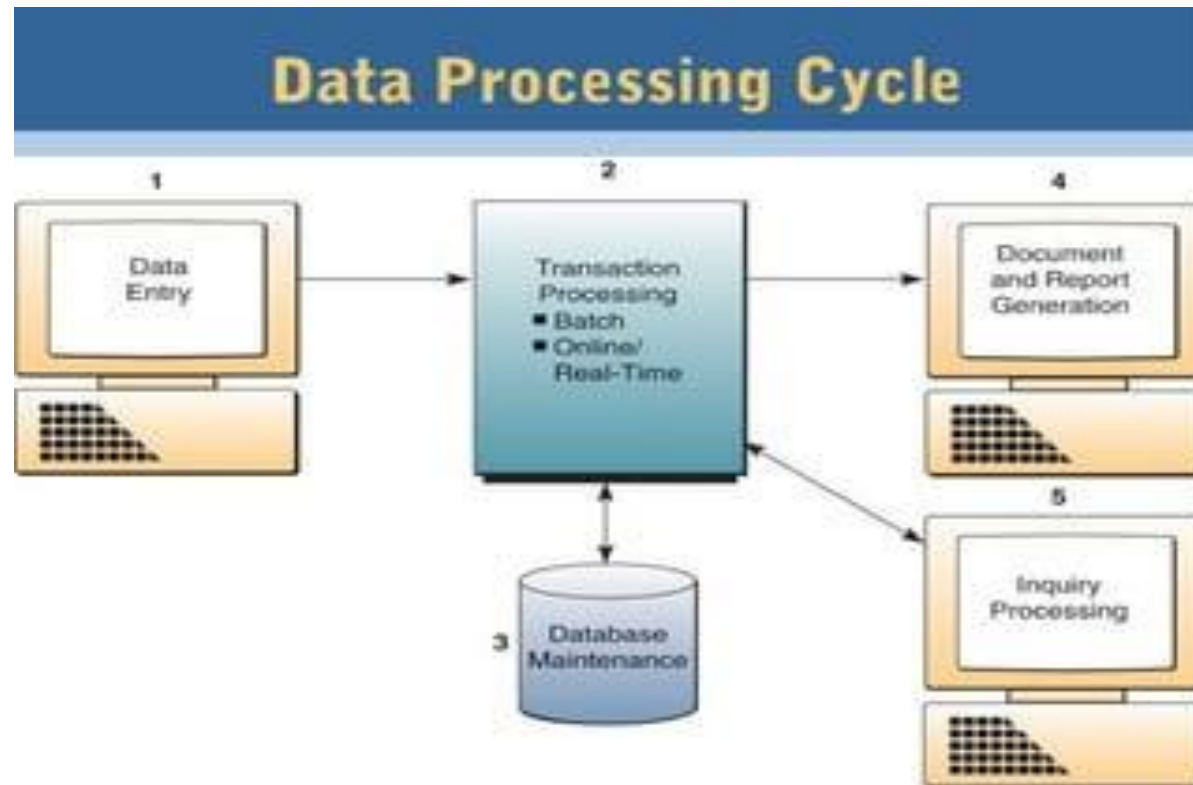
- Transaction Processing System(TPS)
- Office Automation System(OAS)

Operational Information System

- Transaction Processing System(TPS)

A transaction processing system(TPS) is a set of information which processes the data transaction in database system that monitors transaction programs. A TPS is a software system that supports transaction processing.

Operational Information System



Operational Information System

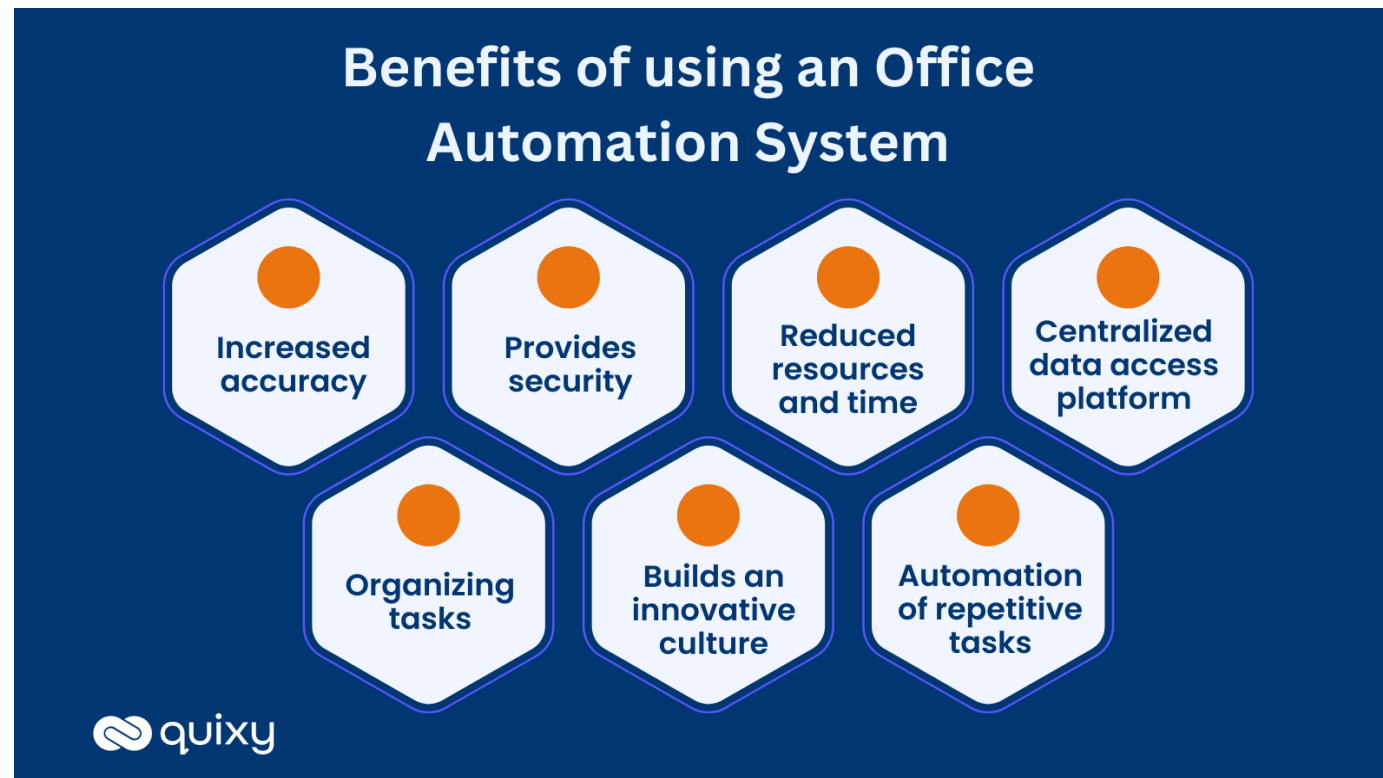
- Office Automation System(OAS)

OAS are computer based information systems that collect,process,store and transmit electronic messages, documents among individuals, work groups and organizations.

Operational Information System



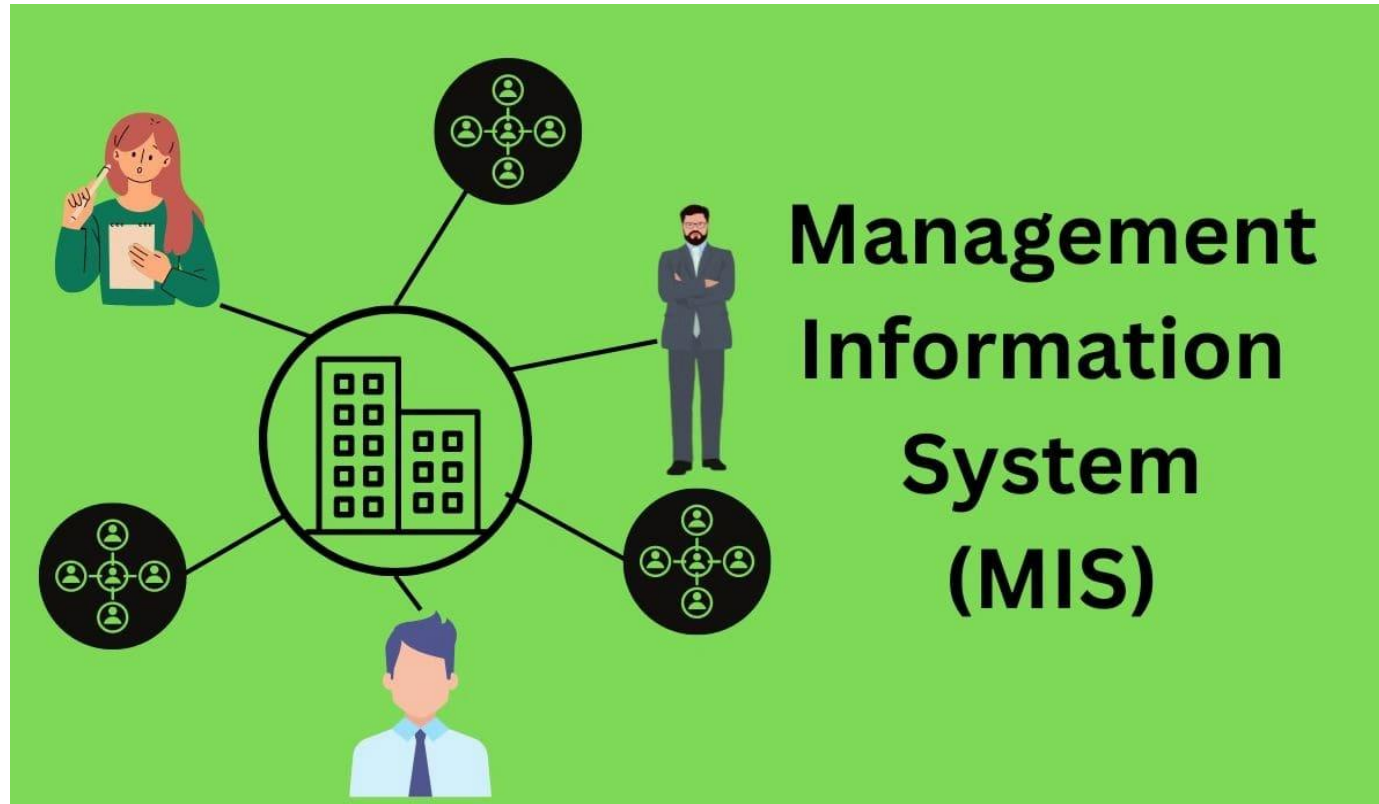
Operational Information System



Tactical or MIS

This type of information system is needed to take short term decision to run the organization efficiently. They assist managers for collecting, storing, managing and retrieving information for organization needed. As for example, statistical summaries, comparative analysis, projections, periodic reports etc.

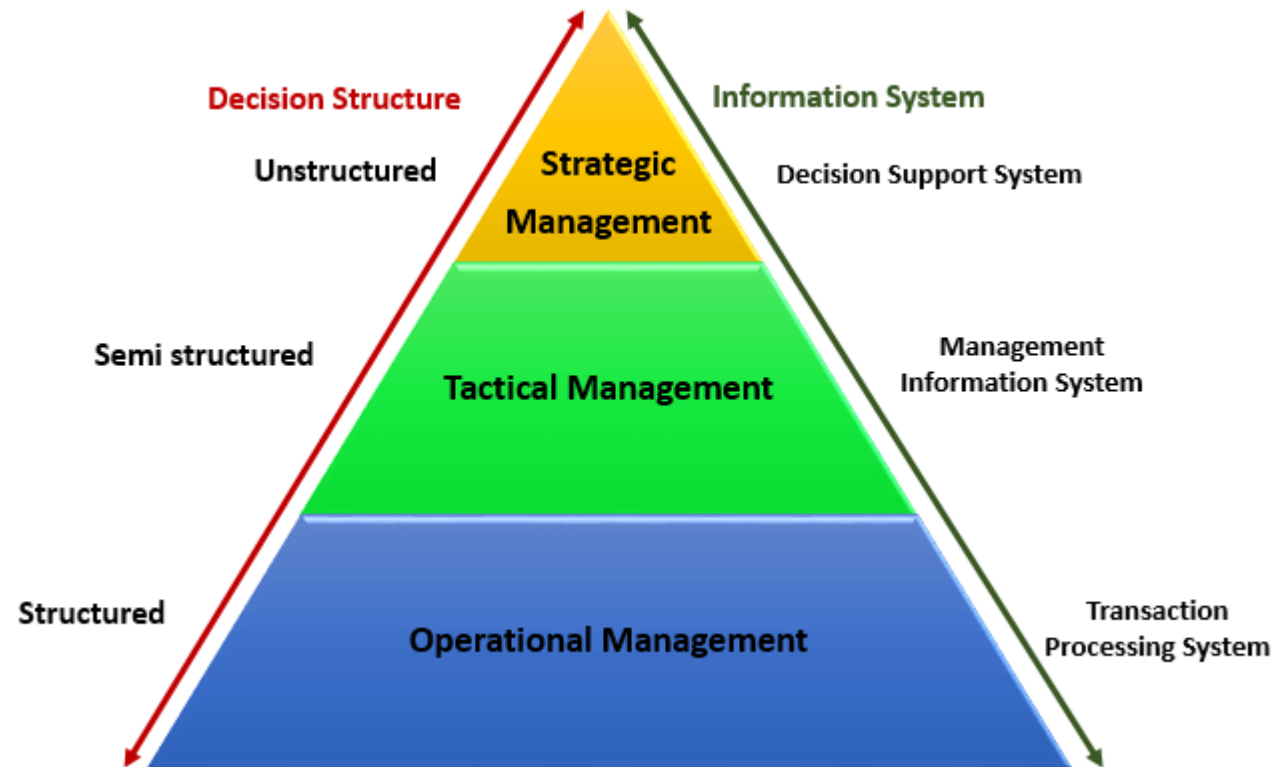
Tactical or MIS



Objectives of MIS

- Data Capturing
- Processing of Data
- Storage
- Retrieval
- Dissemination

Pyramid Structure of IS



Management Information System(MIS)

Management consists of the interlocking functions of creating corporate policy and organizing, planning, controlling and directing an organization's resources in order to achieve the objectives. A **system** is a collection of elements or components that are organized for a common purpose. MIS:

Management Information System(MIS)

MIS provides:

- To the right information
- At the right place
- At the right time
- In the right form
- To the right person

Role of Information System in business Today

The environment in which organizations operate has changed dramatically over the last few years. Increased competition, globalization, the influence of the Internet and international events affect the performance and survival of organizations on a world-wide scale. The Internet has changed the way organizations do business, from the acquisition and servicing of customers to the management of their relations with suppliers. This is not only revolutionizing the way people access information, communicate, shop and entertain themselves but also the way organizations compete and operate.

Role of Information System in business Today

As the cloud continues to gain momentum, companies worldwide are increasing their spending towards the IT sector. Traditional IT systems are declining as modern digital technologies like AI and virtual reality are proving to be strong business driving forces. Indeed, the IT sector is being dominated by emerging cloud infrastructure, AI, VR, block chain and other modern technologies.

The Three Fundamental Roles of Information Systems in Business

- Operations

A business information system carries out specific functions in support of operations, including payroll, employee record storage, preparing and storing company documents and credit card processing. If your company operates a manufacturing line, the information system can schedule tasks and processes while keeping quality records.

The Three Fundamental Roles of Information Systems in Business

Some companies such as graphic designers or advertising companies use the information system to produce their products and services. In supporting operations, the information system can increase efficiency and improve employee productivity.

The Three Fundamental Roles of Information Systems in Business

- Controls

Monitoring and controlling the activities of employees is a core function of information systems. This is especially true of financial transactions in which management has a duty to prevent fraud and theft. In this role, the security of the information system is critical, as managers rely on it to track payments received from customers and invoices from suppliers.

The Three Fundamental Roles of Information Systems in Business

- Decisions

A third fundamental role for information systems is management support in making decisions. Managers can get all the information they need about company activities from the system. Complete, accurate information means management decisions are more effective. More sophisticated systems can go beyond supplying data to running scenarios:

New in Management Information System(MIS)

A **management information system (MIS)** is an information system used for decision-making and for the coordination, control, analysis and visualization of information in an organization.

The study of the management information systems involves people, processes and technology in an organizational context.

New in Management Information System(MIS)

In a corporate setting, the ultimate goal of the use of a management information system is to increase the value and profits of the business. This is done by providing managers with timely and appropriate information allowing them to make effective decisions within a shorter period of time.

History of MIS

While it can be contested that the history of management information systems date as far back as companies using ledgers to keep track of accounting, the modern history of MIS can be divided into five eras originally identified by Kenneth C. Laudon and Jane Laudon in their seminal textbook Management Information Systems.

History of MIS

- First Era – Mainframe and minicomputer computing
- Second Era – Personal computers
- Third Era – Client/server networks
- Fourth Era – Enterprise computing
- Fifth Era – Cloud computing

What Are the Benefits of Globalization?

Globalization impacts businesses in many different ways. But those who decide to take on international expansion find several benefits, including:

1. Access to New Cultures

Globalization makes it easier than ever to access foreign culture, including food, movies, music, and art. This free flow of people, goods, art and information is the reason you can have Thai food delivered to your apartment as you listen to your favorite UK-based artist or stream a Bollywood movie.

What Are the Benefits of Globalization?

2. The Spread of Technology and Innovation

Many countries around the world remain constantly connected, so knowledge and technological advances travel quickly. Because knowledge also transfers so fast, this means that scientific advances made in Asia can be at work in the United States in a matter of days.

What Are the Benefits of Globalization?

3. Lower Costs for Products

Globalization allows companies to find lower-cost ways to produce their products. It also increases global competition which drives prices down and creates a larger variety of choices for consumers. Lowered costs help people in both developing and already-developed countries live better on less money.

What Are the Benefits of Globalization?

4. Higher Standards of Living Across the Globe

Developing nations experience an improved standard of living—thanks to globalization. According to the World Bank , extreme poverty decreased by 35% since 1990. Further, the target of the first Millennium Development Goal was to cut the 1990 poverty rate in half by 2015. This was achieved five years ahead of schedule, in 2010. Across the globe, nearly 1.1 billion people have moved out of extreme poverty since that time.

What Are the Benefits of Globalization?

5. Access to New Markets

Businesses gain a great deal from globalization, including new customers and diverse revenue streams. Companies interested in these benefits look for flexible and innovative ways to grow their business overseas. International Professional Employee Organizations(PEOs) make it easier than ever to employ workers in other countries quickly and compliantly. This means that for many companies, there is no longer the need to establish a foreign entity to expand overseas.

The Challenges of Globalization

While globalization offers many benefits, it's not without challenges. Velocity Global's 2020 state of Global Expansion™ Report: Technology Industry reveals some of the top challenges that U.S. and UK tech leaders face when taking their companies global and leaders of other companies likely face the same obstacles.

The Challenges of Globalization

1. International Recruiting

It's not surprising that 30% of U.S. and UK tech leaders cited international recruiting their most common challenge. Recruiting across borders creates unknowns for HR teams. First, companies create a plan for how they will interview and thoroughly vet candidates to make sure they are qualified when thousands of miles separate them from headquarters. Next, companies need to know the market's demands for salaries and benefits to make competitive offers. To ensure successful hires, HR teams must factor in challenges like time zones, cultural differences and language barriers to find a good fit for the company.

The Challenges of Globalization

2. Managing Employee Immigration

Immigration challenges cause a lot of headaches internally, which is why 28% of U.S. and UK tech leaders agreed it was one of their top challenges. Immigration laws change often, and in some countries, it is extremely difficult to secure visas for employees that are foreign nationals.

The Challenges of Globalization

3. Loss of Cultural Identity

While globalization has made foreign countries easier to access. It has also begun to meld unique societies together. The success of certain cultures throughout the world caused other countries to emulate them. But when cultures begin to lose their distinctive features, we lose our global diversity.

The Challenges of Globalization

4. Foreign Worker Exploitation

Lower costs do benefit many consumers, but it also creates tough competition that leads some companies to search for cheap labor sources. Some western companies ship their production overseas to countries like China and Malaysia.

The Challenges of Globalization

5. Global Expansion Difficulties

For businesses that want to go global and discover the benefits of globalization, setting up a compliant overseas presence is difficult. If companies take the traditional route of setting up an entity, they need substantial upfront capital, sometimes up to \$ 20,000 and costs of \$200,000 annually to maintain the business. Additionally, global businesses must keep up with different and ever-changing labor laws in new countries. When expanding into new countries, companies must be aware of how to navigate new legal systems.

The Challenges of Globalization

6. Immigration Challenges and Local Job Loss

The political climates in the United States and Europe show that there are different viewpoints on the results of globalization. Many countries around the globe are tightening their immigration rules, and it is harder for immigrants to find jobs in new countries.

Globalization Challenges and Opportunities:

Dimensions of Information System

The **dimensions of information systems** include organizations, management, and **information** technology. The key elements of an organization are its people, structure, business processes, politics and culture. An organization coordinates work through a structured hierarchy and formal standard operating procedures.

Dimensions of Information System



Dimensions of Information System

- **Organizational Dimension**

In organizational dimensions, management understands what is embodied in its information system relating organizational stuff like culture, norms and values, core tasks of organization, hierarchy of organization etc. Management should understand that all decisions that we made depend on the information which is provided by information systems. Organizational dimensions of any organization are easy to understand for manager because it is his line and he is better aware of the culture and norms of his organization.

Dimensions of Information System

- **Management Dimension**

In management dimension of information systems, information systems assist manager to allocate resources, hire staff, motivate staff, realize risky areas and allocate experienced staff of such risky areas. Information system assists manager to see different designs of products and to make new products. Manager use different techniques from information system to reduce the cost of its product like; target costing technique, Organization produces its reports, financial statements and forecasts by the help of information systems.

Dimensions of Information System

- Technological dimension

The **technological dimension** of culture is its capital, its tools and skills, and ways of dealing with the physical environment. It is the interface between humanity and nature. Technology is much a cultural **dimension** as beliefs and patterns of interaction. It is symbolic.

Impact of IT on Organizations

Communication:

- E-mail, websites, video conferencing etc.

Information management:

- Data mining of customer information to produce lists for targeted advertising.
- Improved stock control, resulting in less wastage, better cash flow etc.

Security:

- Encryption methods can keep data safe from unauthorized people, both while it is being stored and while it is being sent electronically.

Importance of Information Systems to Society

- Decision-Making
- Security
- Communication
- Productivity
- Added Efficiency
- Analytics
- Communication and Collaboration
- Business